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Tab 1
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Classification Model

In order to predict displacement risk, multiple classification models, such as, Logistic Regression, Random Forest, K-Nearest Neighbors (KNN), and XGBoost models, were developed and evaluated. The main approach focused on balancing the model's predictive performance along with its' ability to accurately classify high-risk cases, with crucial emphasis on recall due to the importance of minimizing the amount of missed displacement cases.

Model Development Approach

The modeling process began with a Logistic Regression as a baseline mode. The initial model uses all the available features, excluding the target variable, the ID variable, and housing_cost_burden to avoid redundancy and inflation. To address class imbalance, SMOTE and class weights were applied. Cross-validation results showed strong performance of a mean F1 score of about 0.73 and a recall score of around 0.84. Different thresholds were tested out, but threshold tuning showed that a lower decision threshold, specifically at 0.3, significantly improved recall at about 0.90, which made it more fitting for classification of high-risk cases.

Two more Logistic Regression models were made using reduced feature sets. The first